

EXP-3

EDA – DATA CLEANING

AIM

To handle missing values, remove duplicate records, convert data types, maintain data consistency, and normalize numerical data.

PROGRAM

```
import pandas as pd

import numpy as np

from sklearn.preprocessing import MinMaxScaler, StandardScaler

data = {
    'Gender': ['Male', 'Female', 'Male', None, 'Female', 'Male', 'Male'],
    'Married': ['Yes', 'No', 'Yes', 'Yes', None, 'No', 'Yes'],
    'Dependents': ['0', '1', '3+', None, '2', '0', '0'],
    'Education': ['Graduate', 'Graduate', 'Not Graduate', 'Graduate', 'Graduate', 'Not Graduate',
'Graduate'],
    'Self_Employed': ['No', 'Yes', None, 'No', 'No', 'Yes', 'No'],
    'ApplicantIncome': [5720, 3076, 5000, 2340, 3276, 4500, 5720],
    'CoapplicantIncome': [0, 1500, 2000, 2546, 0, 1000, 0],
    'LoanAmount': [110, 126, None, 100, 78, 120, 110],
    'Credit_History': [1, 1, None, 0, 1, 1, 1],
    'Property_Area': ['Urban', 'Urban', 'Urban', 'Urban', 'Urban', 'Rural', 'Urban']
}

df = pd.DataFrame(data)

print("Initial Data")
```

```
print(df)
```

```
print("\nMissing Values")
```

```
print(df.isnull().sum())
```

```
for col in ['Gender', 'Married', 'Dependents', 'Self_Employed']:
```

```
    df[col] = df[col].fillna(df[col].mode()[0])
```

```
df['LoanAmount'] = df['LoanAmount'].fillna(df['LoanAmount'].median())
```

```
df['Credit_History'] = df['Credit_History'].fillna(
```

```
    df['Credit_History'].mode()[0]
```

```
)
```

```
df.drop_duplicates(inplace=True)
```

```
df['Dependents'] = df['Dependents'].replace('3+', 3)
```

```
df['Dependents'] = df['Dependents'].astype(int)
```

```
for col in ['Gender', 'Married', 'Education', 'Self_Employed', 'Property_Area']:
```

```
    df[col] = df[col].str.strip().str.capitalize()
```

```
min_max_scaler = MinMaxScaler()
```

```
scale_cols = [
```

```
    'ApplicantIncome',
```

```
    'CoapplicantIncome',
```

```

        'LoanAmount'
    ]

df[scale_cols] = min_max_scaler.fit_transform(df[scale_cols])

standard_scaler = StandardScaler()

df[['Credit_History']] = standard_scaler.fit_transform(
    df[['Credit_History']]
)

print("\nCleaned Data")

print(df)

print("\nFinal Data Information")

print(df.info())

```

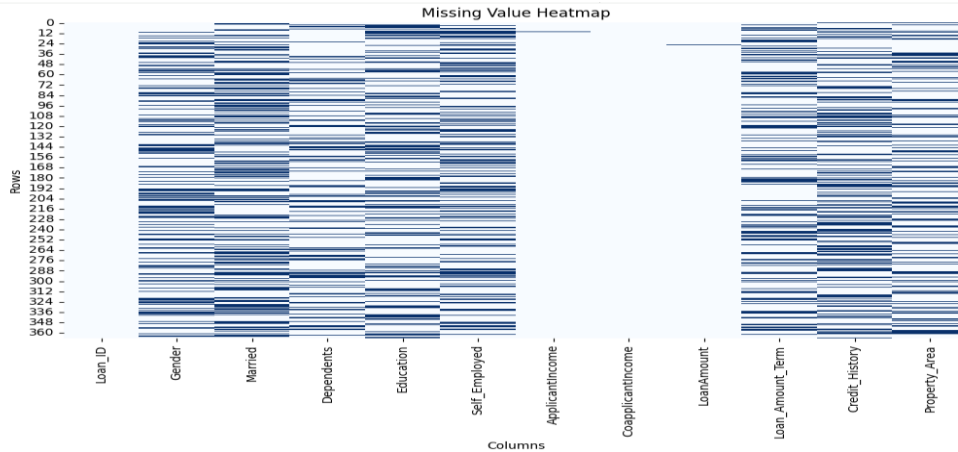
OUTPUT

```

Initial Data Overview:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 367 entries, 0 to 366
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   Loan_ID                367 non-null   object
1   Gender                 249 non-null   object
2   Married                225 non-null   object
3   Dependents             280 non-null   object
4   Education              248 non-null   object
5   Self_Employed          244 non-null   object
6   ApplicantIncome         366 non-null   float64
7   CoapplicantIncome       367 non-null   float64
8   LoanAmount             366 non-null   float64
9   Loan_Amount_Term        255 non-null   object
10  Credit_History          234 non-null   object
11  Property_Area           260 non-null   object
dtypes: float64(3), object(9)
memory usage: 34.5+ KB
None

Missing Values in Each Column:
Loan_ID           0
Gender            118
Married           142
Dependents        87
Education         119
Self_Employed     123
ApplicantIncome    1
CoapplicantIncome  0
LoanAmount         1
Loan_Amount_Term  112
Credit_History    133
Property_Area     107
dtype: int64

```



Removed 0 duplicate rows.

```
Credit_History:
Credit_History
0    -0.641513
1     1.558816
2     1.558816
3    -0.641513
4     1.558816
```

Cleaned Data:

```
Loan_ID  Gender  Married  Dependents  Education  Self_Employed  \
0  LP100000  Female    No           2    Graduate          No
1  LP100001  Female    Yes           0  Not graduate          No
2  LP100002  Male     No           1      None           No
3  LP100003  Male     Yes           3      None           No
4  LP100004  Female    Yes           1      None           No
```

```
ApplicantIncome  CoapplicantIncome  LoanAmount  Loan_Amount_Term  \
0      0.795461      0.461925      0.449799          120
1      0.004022      0.950400      0.550201          180
2      0.706263      0.949907      0.811245          180
3      0.412812      0.819050      0.248996          360
4      0.738437      0.462679      0.550201          360
```

```
Credit_History  Property_Area
0    -0.641513      None
1     1.558816    Urban
2     1.558816    None
3    -0.641513    Rural
4     1.558816    Rural
```

Final Data Information:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 367 entries, 0 to 366
Data columns (total 12 columns):
```

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	Loan_ID	367 non-null	object
1	Gender	367 non-null	object
2	Married	367 non-null	object
3	Dependents	367 non-null	int64
4	Education	248 non-null	object
5	Self_Employed	367 non-null	object
6	ApplicantIncome	366 non-null	float64
7	CoapplicantIncome	367 non-null	float64
8	LoanAmount	367 non-null	float64
9	Loan_Amount_Term	367 non-null	int64
10	Credit_History	367 non-null	float64
11	Property_Area	260 non-null	object

dtypes: float64(4), int64(2), object(6)

memory usage: 34.5+ KB

None

RESULT

Thus, the program was successfully executed. Missing values were handled, duplicate records were removed, data types were converted, categorical values were standardized, and numerical data was successfully normalized.